I Ran Bots Inside Token Factories. Here Is What They Say About PUMP.

Franklin Crypto Research Competition — UBC 2026 · Track: Short, market cap > $1B · Asset: PUMP (pump.fun) · Jean-Baptiste Durand, solo · [School / university group] · contact: [email] · October 2026

Disclosure. From September 23 to October 13, 2026 I ran small automated trading bots on launchpad pools on four chains with $56–122 of external capital; earlier September tests on Pons lost about 0.26 ETH (≈ $700); overall I made about zero (Appendix C). I hold no position in PUMP or SOL. AI use: an assistant checked my arithmetic, produced the figures from my data files and tidied my English; the collector, the bots, the measurements, the judgement and the words are mine.

Terms. Launchpad: a site where anyone mints a token on a bonding curve (a formula that prices the token as people buy and sell); graduation: the token moves to the PumpSwap exchange. Fees: what traders pay; revenue: what pump.fun keeps; take rate: revenue ÷ fees; run-rate: the last 30 days annualised. Factory: an operator that mints pools by the hundred and drains them; family: pools sharing one contract bytecode; drain: the operator pulls the liquidity or sells into the buyer. Address: one wallet (co-signing merges were negligible). Strict P&L: unsold tokens counted at zero. Second buyer: a buy from outside the creator's wallets within 60 seconds. Perpetual: a futures contract with no expiry; funding: its periodic payment between longs and shorts; open interest: the total outstanding. FDV: price × all tokens, circulating or not. Catastrophe stop: the exit if the PUMP/SOL ratio doubles.

Main report: 5 pages (executive summary = page 1). Title page, references and appendices are outside the page count. Figures 1–5 and Table 1 are in the main text.

Executive summary

In one line. I ran bots with real money inside launchpad factories on Arc and Robinhood Chain and came out ahead. Pointed at pump.fun, the same instrument finds a crowd that loses every day and is leaving, a price that pays for one peak week, and a programmed buyer with a date. Short PUMP against SOL, target $0.0022.

What I did. From September 23 to October 13 I ran small buying bots on four chains, from 53 addresses, with about a hundred dollars of outside capital. The operators adapted within two days: none of my first 134 buys was drained, then 10 of 16 were. I swept everything to one public vault (Appendix C). Once a September loss on Pons is counted I ended about even, which validates the instrument rather than the profit. Rebuilt for pump.fun's curve, the same instrument then read sixteen days of pump.fun, position by position.

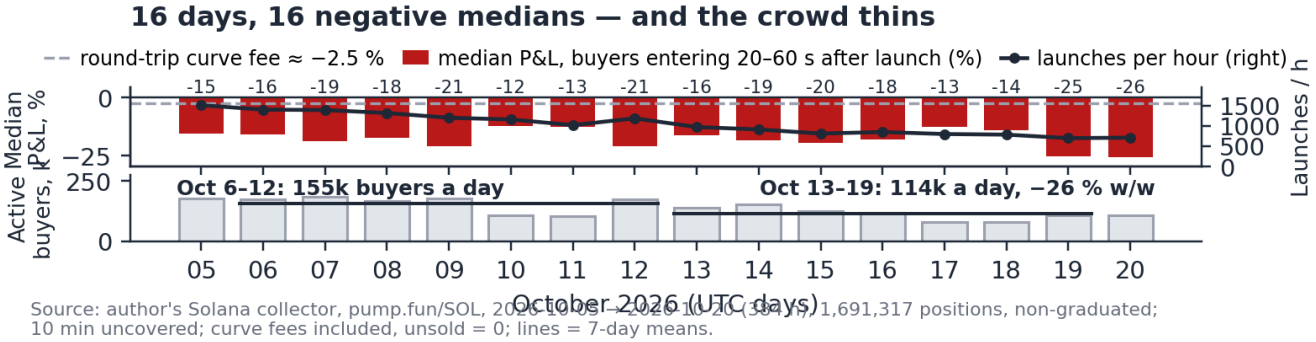
PUMP — SHORT · Price $0.00474 (Oct 21, 2026, ref. 2) · Target $0.0022 (–54 %) · Market cap $2.19B = 2.6× run-rate revenue (last 30 days annualised) · 50 % of revenue goes to buy-back-and-burn under a one-year policy, changeable by one address · Trade: short PUMP perpetual / long SOL, same notional, no leverage, 1 % of the book · Horizon: April 2027 · Thesis stop: weekly revenue ≥ $16M for 4 weeks, buyers > 155k/day for 2 weeks, or launchpad fees ≥ $58M/wk · Catastrophe stop: PUMP/SOL ×2 on a daily close

#1 — The crowd loses, and it is leaving. Over sixteen days in October my collector read every position taken on pump.fun's bonding curve. The crowd proper, the buyers who come in twenty to sixty seconds after a launch, lost a median **17.4 %** per position, where a round trip costs 2.5 % in fees. Every entry window, from the first seconds to five minutes in, is negative, and so is every one of the sixteen days. And the crowd is leaving. Active buyers fell **26 %** from one week to the next, down on all seven same-weekday pairs, and LaunchLab, the other Solana launchpad my collector listens to, lost 71 % of its swaps on the same days (Fig. 1).

#2 — Priced for one peak week. Before the summer mania pump.fun earned about **$7.7M a week** (February to June). The peak week reached **$17.9M**, and the last full week printed **$14.0M**, the third weekly decline in a row. Twice before, revenue reverted after a peak: **−87 %** in six months, **−52 %** in nine (Fig. 4). At today's multiple of 2.6× revenue, the pre-mania base is worth **$0.0022** a token. Part of the toll was given away on the way up, so each fee dollar now carries less to the token than it did at launch (page 3).

#3 — The only programmed buyer has a date. Half of revenue goes to buying back and burning PUMP under a one-year policy that started on April 28, 2026 (ref. 6). Renewal is not committed, and one address can change the share with no timelock (ref. 7). The programme has bought **$490.7M** of PUMP so far, about a quarter of today's market cap, and it is the only programmed buyer. Against it, **6.875B tokens a month** unlock to the team and investors, who paid little or nothing for them (ref. 8). Over the last month the unlocks were already larger than the buybacks: a net yield of **−1.7 %** before the policy even ends.

What would prove me wrong. Weekly revenue at or above **$16M**, what today's price needs, for four weeks in a row, or active buyers back above **155,000** a day for two weeks, or fees across all launchpads back above **$58M** a week, two thirds of the September peak (Fig. D3). A week at $12–14M is not a stop: it values PUMP below today's price.



Source: author's Solana collector, pump.fun/SOL, 2026-10-05 → 2026-10-20 (384 s), 1,691,317 positions, non-graduated; 10 min uncovered; curve fees included, unsold = 0; lines = 7-day means.

Fig. 1 — Sixteen days, sixteen negative medians. Top: median P&L of the crowd's positions and launches per hour; bottom: active buying addresses, with weekly means. Source: my Solana collector, 1.69M positions; coverage in Appendix A.

Project overview (1): how I worked

I had the instrument before I had a thesis. I wrote the rules down before measuring: the pre-registration of September 11 is hashed, and the October rules are dated to the minute. The first time I did it the other way round, a replay of one factory family said +198 %. Priced at the on-chain state of the moment, it said −18.9 % (Appendix A).

The factories were no secret. Most new pools I tracked on those chains were built to be drained. What I did not know was whether bytecode families were real operators, and whether the replay held live. The replay said one recipe survived: buy $0.10–0.20 of a pool 20–45 seconds after birth and sell 110–240 seconds later, before the drain. I ran it on four chains for three weeks (page 1): **+$682 to +$748** on the campaign, about zero overall once the September tests are counted (Appendix C). The replay matched the money: $53.5 a day replayed against $53.1 a day earned, over roughly five hundred round trips.

What I learned is how a factory defends itself, and how fast (Fig. 2). From the evening of October 4, with nothing changed on my side, the share of my buys drained before I could sell went from **none of 134 to 22 % to 10 of 16**, and the whole Robinhood Chain family stopped launching at 18:44 UTC on October 6 and came back three days later. The same night, in the pools where I did not buy, none of 166 third-party buys was drained. That is the control. The rule is a stake wall. Third-party buys of a quarter dollar or more were drained within seconds 156 times out of 157, and my ten-cent buys almost never. On Arc a week later, pools of the family I was buying died 65 seconds after my buy against 213 without me. The families behave as operating units with written rules. One rule I had written in advance failed its own threshold and was not deployed.

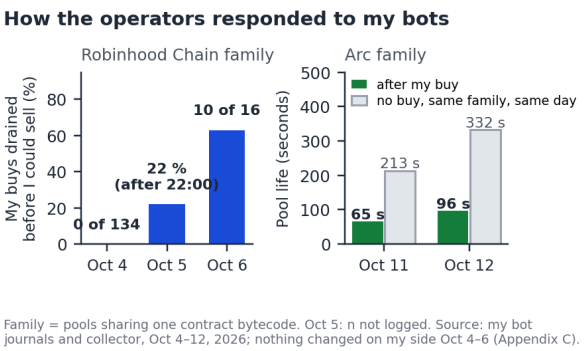


Fig. 2 — The operators' response to my bots: the share of my buys drained before I could sell (one Robinhood Chain family, three consecutive days) and the life of a pool after my buy against pools I left alone (one Arc family). All wallets were swept to one public vault afterwards (Appendix C); the day-by-day picture is Fig. C1.

None of this prices PUMP. Two things carry over. The method holds live, replay against real fills. And the recipe that won on Arc, replayed on pump.fun's exact curve, finds nothing to win. All nine of its cells are negative on each of the sixteen days, with medians between −2.7 % and −7.3 %. On Arc I beat the operators; on pump.fun there is no operator to beat, only a toll, and the crowd pays it. The instrument also passes a known-answer check (a graduating token must multiply ×14.7 on the curve, and it does), and it produces every pump.fun number that follows.

Project overview (2): pump.fun under my instrument

The business. pump.fun earns on four products: the bonding curve; PumpSwap, where graduated tokens trade and which now earns more than the curve ($106M of fees in September against $46M); the Terminal and the mobile app; and, since mid-October, perpetuals routed to Hyperliquid. Third-party terminals such as Axiom ($30.6M of fees in a month) sit between the crowd and the curve. My own crowd numbers cover the SOL-quoted curve. The company, Baton Corporation Ltd, is registered in England (directors Noah Tweedale, Alon Cohen, Dylan Kerler). It raised $1.32B in July 2025 at $0.004 a token. Its filed accounts were overdue at the time of the Blockworks filing, and one address can change the buyback share with no timelock (ref. 7).

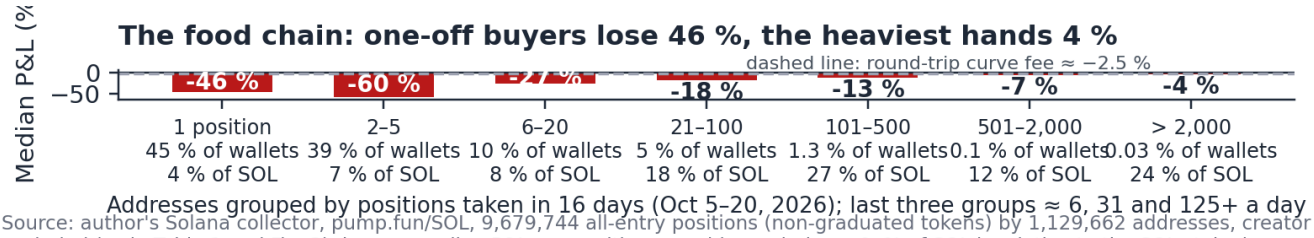
Sixteen days (October 5–20, 2026). 395,253 SOL-quoted tokens were launched, 1,029 an hour. All launch modes together are nearly twice that, and the SOL-quoted family is about a fifth of pump.fun's fees. Half of the tokens found a second buyer within a minute (51 %), a little below Zora, on seven times Zora's throughput, and the proxy counts scripted buyers too. Yet the median position lost 13.3 % across all entry times and **17.4 %** for the crowd proper (strict P&L, or −15 % valued at the last curve price, and negative on all sixteen days under every rule, Appendix A). Creators as a group lost 28.8 % of what they put in. The clearest operator signature is a creator script that sells within seconds of an outside buy 1.8 to 4.6 times more often than at matched moments.

Venue (chain)	Launches per hour	Second buyer ≤ 60 s	Crowd median P&L	Operator pull or sell within seconds of an outside buy (× matched control)	Inorganic signature
pump.fun (Solana), 16 days	**1,029**	**51 %**	**–17.4 %**	**×1.8–4.6 (×2.1 controlled)**	**12 % never trade; buyers –26 % w/w**
PumpSwap (post-graduation, 19 h)	36–51 playable pools	100 %	–20.5 % night / +1.7 % day	none	virtual reserve counted in price
LaunchLab (Raydium, other quote tokens)	77	93 %	–23.2 %	×10.4	StonkFun = 95 % of September launches
Zora	138	56 %	–0.6 to –2 %	none (0 / 606)	one operator = 83 % of births
Pons V2	354	n/a	–5.9 % (3rd-party)	5 of 6 drained in 1.4 s	fees –43 % after gas subsidy ended

Table 1 — Five venues under the same scanner. The busiest launchpad still costs the crowd 17 % per position. Read signatures, not levels: the other venues are snapshots of a few hours to a day (Oct 4–8). Source: my collector; columns in Appendix B.

The crowd is thinning. Active buying addresses fell from 155,000 a day in the first full week to 114,000 in the second, **−26 %**, and the fall shows on each of the seven same-weekday pairs while SOL itself lost 8 %. Creations, graduations, buyers and revenue all fell on every pair, and PumpSwap, the part I do not measure, is 30 % below its peak week. A week down 14 % is not rare in the public series (7 of the last 40). What is new is the breadth: all seven pairs, new entrants −21 %, and every launch mode together −24 % in positions. On the same days LaunchLab, the other Solana launchpad the collector listens to, lost 71 % of its swaps. The game is cooling across launchpads.

The food chain, hand by hand (Fig. 3). I grouped the 1.13 million buying addresses by the number of positions they took in the sixteen days. The 45 % with a single position lose a median **46 %**. The 395 addresses with more than two thousand positions lose **4 %**, little more than the fee. Nobody wins on the median. The crowd and the creators together lost more in those sixteen days than pump.fun earned in revenue. The 84 % of addresses with five positions or fewer place 11 % of the SOL staked on the curve, and the 1.4 % with more than a hundred positions place **62 %**. The heaviest hands lose least, and the only sure winner is the fee.



Source: author's Solana collector, pump.fun/SOL, 9,679,744 all-entry positions (non-graduated tokens) by 1,129,662 addresses, creator excluded (as in Table 1 and Fig. 3); bars = median P&L per position, unsold remainder = 0. % of SOL is relative to the SOL staked on non-graduated tokens; addresses with more than 100 positions place 62 % of it.

Fig. 3 — The food chain: median P&L per position, by the number of positions an address took in the sixteen days. Source: my Solana collector, 9.68M positions on non-graduated tokens, strict P&L.

Three public facts (Figs. D1–D3). *The toll was cut*: fees multiplied by 2.5 this summer but revenue only by 2, because the take rate fell, partly by decision (creator rewards, Project Ascend) and partly by mix (PumpSwap's share grew). A RICO claim against the company survived a motion to dismiss on August 31, 2026 (ref. 11), and UK users have been blocked since a regulator's warning (ref. 12). *Three launchpads, one summer*: Pons peaked first, at the end of August, then lost 93 % of its weekly fees in six weeks. pump.fun peaked a month later and has fallen since. *Share came back as Pons collapsed* (Fig. D3): across the launchpads alone, weekly fees went $6.5M → $85M → $26M (June, early September, mid-October) while pump.fun's share went 83 % → 10 % → 52 %.

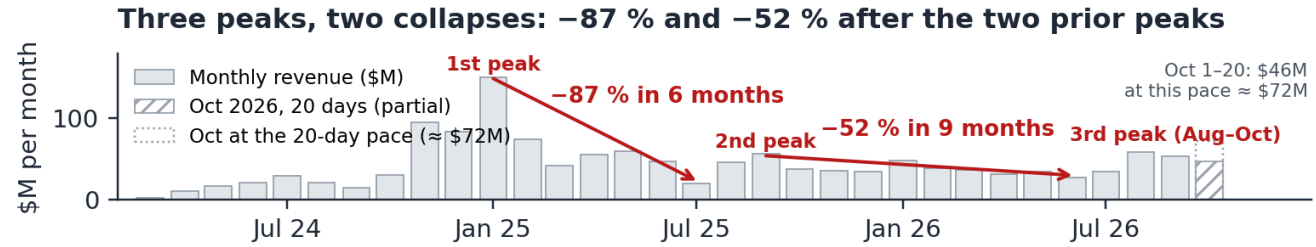
Investment proposal (1): what it is worth

Comparables (market cap / annualised 30-day revenue, October 21). PUMP **2.6**× (1.6× to 4.2× over the last year) · PONS 3.2× · RAY 10× · JUP 18.3× · HYPE 22×. PUMP is the cheapest of the group even on cash returned to holders. The discount is for durability, and I agree with it.

Target, two methods. *A multiple on base revenue.* The February–June 2026 average, $398M a year, at today's 2.6× gives a market cap of $1.04B, or **$0.0022 a token (−54 %)**. The multiple is pro-cyclical (it was 1.9× at the

June trough), so 2.6× on base revenue is generous. On 90-day revenue the market pays 3.2×, which would give $0.0028, the top of the range. I take the low end because unlocks exceed buybacks at base revenue and the buyback policy ends in April. *A buyback-coverage price*, a stress case with no multiple: the price at which half of revenue absorbs a year of unlocks is $0.0051 at the peak run-rate, within 7 % of today, and **$0.0024** at base revenue. Both methods are linear in revenue, so their agreement proves little.

Revenue base ↓ / multiple →	1.9× (June 2026)	2.6× (today, 30-day basis)	3.2× (today on a 90-day basis; PONS today)
June-2026 trough, $319M/yr	$0.0013	$0.0018	$0.0022
Feb–Jun 2026 average, $398M/yr	$0.0016	**$0.0022**	$0.0028
Trailing 12 months, $465M/yr	$0.0019	$0.0026	$0.0032
Last full week, $728M/yr	$0.0030	$0.0041	$0.0050
Peak run-rate, $836M/yr	$0.0034	$0.0047 (today)	$0.0058



Source: DefiLlama dailyRevenue, pump.fun parent protocol (launchpad, PumpSwap, Terminal, app), Mar 2024 – Oct 20, 2026; October partial (hatched).

Fig. 4 — Three peaks, two collapses: pump.fun monthly revenue since March 2024 (October partial). The weekly series peaked in the last week of September and is −22 % since. Source: DefiLlama, series as of October 21, 2026.

Scenarios (six months). Bear, 25 %: trough revenue at the trough multiple and the policy not renewed, $0.0013. Base, 50 %: $0.0022. Bull, 25 %: the mania stabilises near $15M a week and the market re-rates to 3.25×, $0.0055. If the market paid the 2025 long's 10× on base revenue (ref. 13), PUMP would be worth $0.0086. That is where I would admit I was wrong, and the revenue stop fires long before. The probability-weighted price is about $0.0028 (reverse check in Appendix D).

Net supply, and who holds it. The buyback, worth 19 % of market cap a year at the run-rate and 9 % at base revenue, is the bulls' best argument, and it is dated. Each month 6.875B tokens (about $32.6M) unlock to a team and investors whose cost basis is zero or the sale price (refs 7–8). Insiders kept 92 % of the July cliff (ref. 9), so I count unlocks as a weight on the price rather than a trigger, and the path uses their dates as checkpoints. Over the last month $29.5M was bought back against $32.6M unlocked, a net yield of −1.7 %. The holder map (ref. 4) shows 43 % of the circulating supply in some twenty round-number multisig vaults, held rather than sold, and 14 % on exchanges and the Hyperliquid bridge. The float is about a third.

Investment proposal (2): risks and trade

Tested and rejected (or not settled). *The crowd always lost and pump.fun survived two collapses*: true; my claim is that this crowd is thinning while the price assumes it stays. *The crowd is rotating back to pump.fun*: its share of launchpad fees did rebound, but the pie itself is far smaller than at the peak, buyers are down a quarter and LaunchLab lost more than two thirds of its swaps in the same week. This is one of the stop readings. *Fewer but richer buyers*: revenue per active buyer rose 15 % in the week. On Arc the same pattern preceded the September break, and the heavy hands depend on the occasional ones for their exits. *The volume is wash trading*: 1.3 % of pump.fun volume on the strict lower bound (ref. 15). My claim is the opposite: real, negative-sum volume, which is worse for durability.

Risk	Probability	Effect	Trigger	Response
The crowd returns and the multiple follows	medium	bull case $0.0055, or the stop	revenue ≥ $16M/wk for 4 weeks; buyers > 155k/day for 2 weeks	thesis stop
Take rate restored (it ticked up from 30 % to 37 % weekly since mid-September), buyback raised or topped up from the $1.32B raised (ref. 7)	low–medium	+20–40 %	weekly take rate above 40 %; config update	thesis stop; halve the position
Squeeze: OI 16 % of market cap, ~14 % of circulating supply on exchanges, ×3.6 this summer	medium	pair ratio ×2	listing, buyback news, SOL rally	catastrophe stop; 1 % of the book; no leverage
Funding turns against the short (seen Oct 17–19); product expansion (app, perpetuals) lifts revenue per buyer	medium	–15 to –25 % of notional; revenue holds while buyers fall	7-day funding below –20 %/yr; the monthly prints	halve the position; thesis stop, not crowd numbers

Catalysts, dated — the path (Fig. 5). The weekly prints of November 2, 9, 16 and 23: a week near $14M is worth $0.0041 at a constant multiple, and the first third of the profit is taken there. The November monthly print on December 1, the first full post-peak month. The monthly unlock on the 12th, while revenue drifts toward its base ($0.0028 at the higher multiple, the second third). And the end of the one-year buyback policy, about April 2027, with no commitment to renew ($0.0022 at today's multiple on base revenue, the balance).

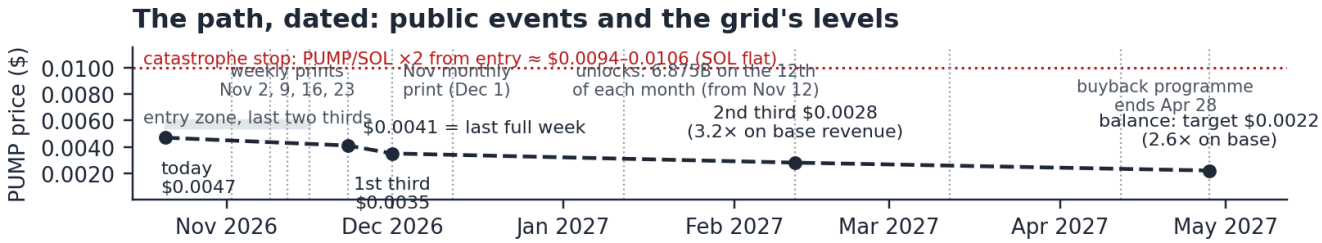


Fig. 5 — The path, dated: public events of the next six months and the grid's levels; grey band = the last two thirds of the entry on a rebound to $0.0055–0.0060 by November 16. Sources: DefiLlama, Blockworks filing, buyback policy.

Trade. A pair: short the PUMP perpetual, long SOL for the same notional, no leverage, 1 % of the book. The Binance perpetual trades $222M a day with $352M of open interest (ref. 3), so a $10M short is a small fraction of it, and funding currently pays the short. Build: a third at $0.0047 now, two thirds on a rebound to $0.0055–0.0060 before the November 16 print, or at market after that print if it confirms (table at $0.0053). At this volatility any price stop within reach is noise, hence a thesis stop on public series (weekly revenue, my buyer counts) and a catastrophe stop at a doubling of the PUMP/SOL ratio on a daily close.

Six-month package per $ of notional	PUMP leg	Package, funding net of SOL leg	Probability
Bear · Base · Bull (PUMP –72 % · –54 % · +16 %)	+75 % · +58 % · –4 %	+76 % · +59 % · –3 %	25 % · 50 % · 25 %
Catastrophe stop (pair ratio ×2)	–100 %	–99 %	12 % under my drift, 38 % with none (rough)

Assumptions: entry $0.0053, SOL leg beta-neutral, funding +8 %/yr for three months. **This is a thesis trade: it pays about +30 % of notional if revenue reverts as it did twice before, and loses about 8 % if nothing moves (rough, biased upward).**

Conclusion

Short PUMP against SOL, to $0.0022, by April 2027. I played the game with my own money, and the instrument that found the factories' rules finds a crowd that loses to the toll and is leaving, a price that pays for a peak week, and a programmed buyer with a date. Four weeks of revenue at or above $16M, or buyers back above 155,000 a day, would change my mind.

References

Primary data (all pulled by the author; raw responses saved in the repository under `data/raw/`): 1. DefiLlama fees & revenue API — `api.llama.fi/summary/fees/pump?dataType=dailyFeesIdailyRevenue` (pump.fun parent: launchpad, PumpSwap, Terminal, mobile) and `summary/fees/{pons, arguspad, flap, …}` for 15 other launchpads; accessed Oct 21, 2026. 2. CoinGecko API — `coins/pump-fun/market_chart`, `coins/solana/market_chart`, `coins/markets` (HYPE, JUP, RAY, PONS); 365-day history, accessed Oct 21, 2026; 24-hour spot volume, Oct 22, 2026. 3. Binance Futures & Spot APIs — `fapi/v1/fundingRate` (PUMPUSDT, 4-hour settlement since Jul 30, 2026), `futures/data/openInterestHist`, `globalLongShortAccountRatio`, `topLongShortPositionRatio`, `premiumIndex`, `ticker/24hr`; Hyperliquid `info` (`metaAndAssetCtxs`); accessed Oct 21–22, 2026. 4. Solscan — PUMP token page, holders tab (top 80 accounts with labels), https://solscan.io/token/pumpCmXqMfrsAkQ5r49WcJnRayYRqmXz6ae8H7H9Dfn ; read Oct 22, 2026. 5. Author's collector, lab tools and bot journals — 35 EVM chains + Solana; methods, pre-registered rules, controls, daily results; GitHub (to be published before submission); vault 0x7d85bF7a82470837A1d832e4fa503a7ebF20ca97 and four sweep transactions on Arc (Oct 7 and 13, 2026), Appendix C.

Issuer and token documentation: 6. pump.fun, "PUMP token" page — buyback programme, cumulative buybacks and burns, https://pump.fun/pump-token ; read Oct 20, 2026. pump.fun fee documentation, https://pump.fun/docs/fees ; updated Oct 17, 2026. 7. Blockworks Token Transparency, filing for PUMP (B1 v1.1, filed Sep 21, 2026) — allocation, ICO tranches, vesting schedule, buyback policy (`buyback_basis_points` = 5,000; global config authority; no timelock; one-year commitment, no commitment to renew), legal entity, https://blockworks.com/token-transparency/filing/pump ; read Oct 22, 2026. 8. Tokenomist, "PUMP tokenomics: what the July 12, 2026 insider cliff means for a fixed 1T supply", https://tokenomist.ai/research/pump-tokenomics-what-the-july-12-2026-insider-cliff-means-for-a-fixed-1t-supply-2 ; DropsTab vesting page, https://dropstab.com/coins/pump-fun/vesting. 9. 4Pillars, "PUMP insiders kept 92 % of the July unlock", https://research.4pillars.io/en/data/content/pump-insiders-kept-92-of-the-july-unlock (Jul–Aug 2026); 4Pillars, "Debunking pump.fun revenue manipulation FUD" (Mar 17, 2026), https://research.4pillars.io/en/research/debunking-pumpfun-revenue-manipulation-fud. 10. CoinDesk, "Pump.fun has been using all its revenue to burn its token. Now it's changing course" (Apr 29, 2026; 36 % of circulating supply burned, 50 % of revenue to buybacks), https://www.coindesk.com/markets/2026/04/29/pump-fun-burns-36-of-pump-supply-in-usd370-million-wipe-locks-50-revenue-into-ongoing-buybacks ; The Defiant, "Pump.fun's PUMP buybacks top $400M as token stays flat", https://thedefiant.io/news/defi/pump-fun-s-pump-buybacks-top-usd400m-as-token-stays-flat.

Legal and regulatory: 11. Protos, "Solana Labs dropped from Burwick Law's pump.fun lawsuit" — *Aguilar v. Baton Corp* (S.D.N.Y., Judge McMahon), RICO claims sustained, Aug 31, 2026, https://protos.com/solana-labs-dropped-from-burwick-laws-pump-fun-lawsuit/. 12. Decrypt, "Solana meme coin launchpad pump.fun bans UK users after regulator warning" (Dec 2024), https://decrypt.co/295355/solana-pump-fun-bans-uk-users-regulator ; Cointelegraph, "FCA issues warning, bans pump.fun in the UK" (Dec 3, 2024), https://cointelegraph.com/news/fca-issues-warning-bans-pump-fun-uk.

The bull case I argue against: 13. Delphi Digital, "Bigger than Binance: how pump.fun's $1.3B raise sets up crypto's most hated rally" (Jul 10, 2025), https://members.delphidigital.io/reports/bigger-than-binance-how-pump-funs-1-3b-raise-sets-up-cryptos-most-hated-rally ; Tang, Hirsch, Killmer (Cornell Blockchain Club), "Pump.fun: the long degen bet", Long > $250M track, Franklin Crypto Research Competition, MBC 2025 — the finalist long I argue against (target $0.0152).

Valuation method references: 14. Token Terminal, "Key metrics FAQ" (fees, revenue, earnings, FDV), https://tokenterminal.com/articles/token-terminal-key-metrics-faq ; TokenIntel, "Buybacks are the mechanism, not the moat" (Jun 2, 2026; net buyback yield), https://tokenintel.org/reports/buybacks-are-the-mechanism-not-the-moat-2026-06-02 ; CF Benchmarks, "Pricing the perp DEX leader: a valuation framework for Hyperliquid" (Jun 10, 2026), https://www.cfbenchmarks.com/blog/pricing-the-perp-dex-leader-a-valuation-framework-for-hyperliquid ; CoinShares, "Hyperliquid primer and 5-year valuation framework" (Jun 2, 2026), https://coinshares.com/us/insights/research-data/hyperliquid-primer-and-5-year-valuation-framework/.

Academic and industry research on pump.fun: 15. Szwajcok, Tsuchiya, Liu, Soska, Payer, Christin, "Meme Coin Factories: Uncovering Large-Scale Manipulations on pump.fun", arXiv 2609.10246, ACM CCS 2026, https://arxiv.org/abs/2609.10246 — wash trading ≥ 17 % of trading transactions (strict lower bound), 1.32 % of volume; graduation 1.02 %. 16. Marino, Naviglio, Tarantelli, Lillo, "Predicting the success of new crypto-tokens: the Pump.fun case", arXiv 2602.14860 (Feb 2026), https://arxiv.org/abs/2602.14860 — graduation 0.63 %, median curve life 4.4 min. 17. Ding, Luo, Xu, "Decompose Market Manipulation Strategies: Evidence from On-chain Meme Coin Market", SBFC 2025, https://sbfc.sydney.edu.au/2025/papers/SBFC2025_4G3_P154.pdf — sniper and "rat" bots; creators as the most frequent dumpers. 18. Kamat, "Do Two On-Chain Observation Pipelines See the Same Tokens?", arXiv 2609.18975 (Sep 2026), https://arxiv.org/abs/2609.18975 — why collector coverage is reported.

Appendices (outside the 5-page count)

Reference material — definitions, the confidence table, the reconciliation of the money, the valuation arithmetic. The main report stands without it.

Glossary. *Hand / address*: one wallet (merging co-signing wallets changed nothing in this data); not a person. *Strict P&L*: profit or loss per position with tokens still unsold counted at zero. *Positive control*: a case whose answer is known in advance (a token that graduates must multiply ×14.7 on the curve); if the method finds it, the method works. *Second buyer*: share of new tokens that get a buyer from outside the creator's wallets within 60 seconds — a rough sign of real demand. *Take rate*: the share of trading fees pump.fun keeps as revenue. *Bonding curve / graduation*: the formula that prices a new token as people buy and sell; graduation moves the token to the PumpSwap exchange. *Perpetual, open interest, funding*: a futures contract with no expiry; the total outstanding; the periodic payment between longs and shorts. *FDV vs market cap*: all tokens that will ever exist, vs those in circulation. *Constant multiple*: assuming the price-to-revenue ratio stays where it is today.

Appendix A — The instrument

Collector. A read-only collector on 35 EVM chains plus Solana. It decodes launchpad events (bonding-curve buys and sells, pool creations, graduations, mints), stores them as hourly JSONL, and enriches each record with the real sender of the transaction (`tx.from` , not the router in the event), the deployer, and the size and simhash of the token's bytecode. Per-address P&L is reconstructed on bought tokens only (sales of tokens never bought — allocations, transfers — are excluded; this one rule removed several "winners" at several "winners" at absurd returns). Every test started with a written rule (thresholds, same-day controls, what would count as a positive result) and ended with a verdict against it; a rate-limited RPC returns "zero logs" without error, so every zero was challenged with an unfiltered canary request.

Known pitfalls, documented as they happened (each cost me a wrong conclusion): quoting a token at the last logged price instead of its historical on-chain state (a +198 % replay that was −18.9 % for real); a constant-liquidity simulator (off by ×30 on thin pools); the router counted as the buyer; dead pools left in a replay (17 % of them); an unmatched control; journals that count one side only — only on-chain balances are trusted for money.

Windows. pump.fun: 16 full UTC days (Oct 5–20, 2026, 384 h), ≈ 65.3M transactions, 395,253 launches; coverage 23,030 of 23,040 minutes, canary 99.4 % on average (98.3 % worst day); one run per day, same parameters, every daily result published. Arc factory pools: 17 days (Oct 4–21). Robinhood Chain family: Oct 4–9. Other venues in Table 1: 6–22 h each (Oct 4–8).

Confidence — one line per proprietary number.

Number	n	Window (UTC)	Method	95 % CI	Test	p
Crowd median P&L, entry 20–60 s, non-graduated tokens: **–17.4 %** (valued at the last curve price –15.4 %; excluding the 11.2 % of positions never sold –12.2 %; closed positions only –11.6 %)	1,691,317 positions	Oct 5–20, 16 days	strict P&L on curve flows, curve fees included, unsold remainder = 0; each variant negative on 16/16 days (`results/robustness_2060.csv`)	[–17.5 %; –17.3 %] (order statistics, positions treated as independent); [–19.8 %; –16.0 %] = 95 % CI of the mean of the 16 daily medians (–17.9 %), by bootstrap	sign test, 16/16 days negative (days of one regime are not independent: indicative)	1.5×10^{-5}
Same, mean –24.3 %; losing positions 74.8 %	1,691,317	same	arithmetic mean; count	±0.1 %	—	—
Hour by hour: 384/384 hours negative (exact-curve replay, E = 60 s, H = 60 s)	384 hours	same	median per hour of birth	—	count only (hours of one regime are not independent)	—
Out of sample: replay cell chosen on the first half of each day, judged on the second	16 days	same	forward walk, split at mid-day	—	second half negative on 16/16 days (–6.9 % to –8.0 %)	—
Positive control: graduated tokens ×14.70 birth → end of curve	9,916 graduated	same	theory $(115/30)^2 = 14.7$	—	—	—
Creator sells ≤ 3 s after an outside buy: ×1.8–4.6 vs matched control (×2.1 once conditioned on prior third-party sales, 19-h sub-window)	2,780,579 outside buys	per day	matched control: same tokens, same age, no outside buy	—	ratio > 1 on 16/16 days	—
Active buyers –26 % week over week (155,300 → 114,481 per day)	7 + 7 days	Oct 6–12 vs Oct 13–19	distinct active buying addresses with ≥ 1 curve buy that day	—	same-weekday comparison; creations –33 % and revenue –15 % move the same way; SOL –8 % is context, not a control	—
Factory pools tracked on Arc: 93 % of trading addresses end negative	7,160 addresses, 1.54M trades	Oct 4–21, 408 h	realised-or-written-off (open positions at last curve price, zero if the pool was drained)	92.9 % ± 0.6 pt (binomial)	no regular winner (≥ $20/day in two of three 136-h windows)	—

Caveats: the 16 days are one market regime (post-peak October 2026) and consecutive days are not independent, so the day-level sign test is indicative, not a proof of generality; the daily medians (–12 % to –26 %) are the honest spread. Kamat (2026, ref. 18) shows that two collectors can see almost disjoint token sets, which is why coverage is reported. Addresses are wallets; the food-chain script's co-signer grouping returned the same count (1,129,662), so no merge is applied.

What my method misses. Sales after graduation on PumpSwap are invisible, so P&L on graduated tokens is a lower bound. The strict rule (unsold tokens at zero) overstates the loss on tokens still alive at the end of the window: valued at the last curve price the median is –15.4 %, and –12.2 % excluding never-sold positions; the last two days carry twice the usual share of never-sold positions (18–21 % against 5–15 %), which drives their strict medians to –25 % while the rest of their positions sit at –13 % to –15 %, in line with the other days; the observation horizon is the same every day (each day is followed to 00:00 UTC the next), so this is tokens ceasing to trade, not censoring; across days the never-sold share tracks the strict median (r = –0.89) but not the median of the rest (r = –0.5). Bots and wash trades are not filtered out of the crowd; if anything they make its median *less* negative. An address is not a person. Solana transaction, priority and trading-bot fees are not deducted, so the crowd's true P&L is worse than shown; on EVM chains gas is not deducted and 4.7 % of Arc senders are unattributed. My graduation rate (2.5 %) is above the literature's 0.6–1.0 % (refs 15–16); a mania month and a different perimeter explain part of the gap, I do not fully explain it, and I cite both.

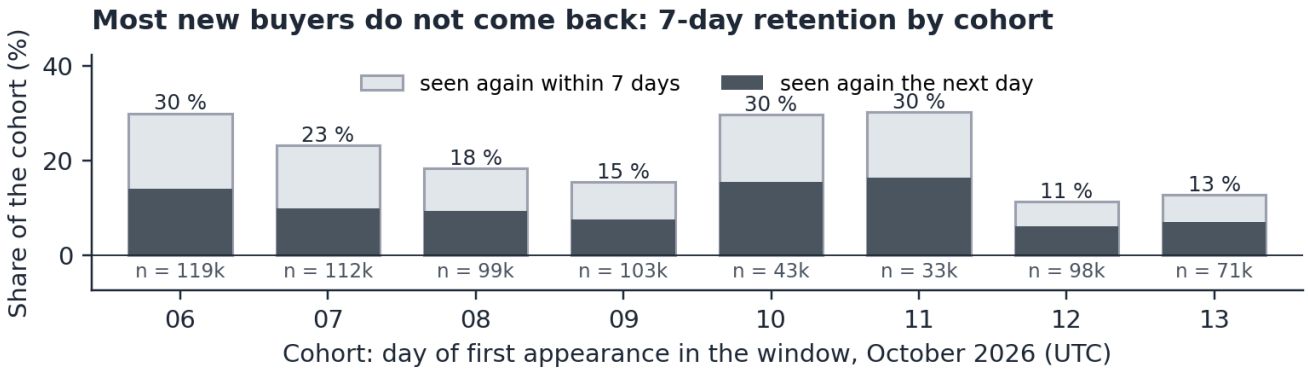
Appendix B — Five launchpads under the same instrument

Columns of Table 1. *Launches/h*: new tokens or pools per hour. *Second buyer*: share of new tokens with at least one buy within 60 s from an address outside the creator's cluster — my proxy for organic demand. *Crowd median P&L*: for pump.fun, the strict P&L of entries 20–60 s after launch; for the other venues, third-party buys within 600 s of birth (gas not deducted). *Operator pull or sell after a third-party buy*: how much more often the operator pulls liquidity (EVM venues) or the creator sells (pump.fun) within seconds of an outside buy than at matched moments on the same tokens at the same age (pump.fun ×1.8–4.6 by day; ×2.1 once conditioned on prior third-party sales, 19-hour sub-window — the raw ratio overstates targeting). *Inorganic signature*: the clearest sign of manufactured activity on that venue. P&L rules differ by venue — pump.fun strict (unsold remainder = 0), Arc realised-or-written-off (open positions at the last curve price, zero if drained) — so the table compares signatures, not levels.

Factory pools I tracked on Arc, 17 days (Oct 4–21). 7,160 trading addresses; 508 end positive (+$38.1k) and 6,652 negative (−$46.1M); 1.54M trades; no regular winner (≥ $20/day in two of three 136-hour windows); median crowd position $146 (118-h sub-window, Oct 4–9). The 7,160 exclude 11,680 liquidity addresses (deployers and drains); the dollar sum (mean loss $6,930 against a median of $47) is probably mostly the factories' own capital cycling through their pools, which is why I cite the count and the median, not the sum.

Same-chain control. LaunchLab (Raydium), the other Solana launchpad the collector listens to: swaps per day 122,640 (Oct 6–12) → 35,484 (Oct 13–19), −71 %, against −26 % for pump.fun's active buyers (COMPUTED from the daily `verificationsLaunchLab.echanges`).

pump.fun, 16 days (Oct 5–20). 1,029 launches/h for SOL-quoted tokens (1,506/h on Oct 5 → 705/h on Oct 20; all launch modes 1,824/h, 700,261 tokens, 2.08 % graduated; the SOL-quoted family is 59.6 % of curve swaps); 51.3 % second buyer (44.6 % strict); exact-curve replay at E = 60 s, H = 60 s: median −2.5 % (the round-trip fee) on each of the 16 days; at the recipe's own cells (entries 20–45 s, exits 120–240 s later) the median is −2.7 % to −7.3 %, negative 16/16 days in each cell; after-graduation entries on PumpSwap (5–45 s, 19-h sub-window only): median −20.5 % at night, +1.7 % by day. Users: active buying addresses 155,300/day (Oct 6–12) → 114,481/day (Oct 13–19), −26 % while SOL fell 8 %; new entrants not seen in the previous 7 days 72,409 (Tue Oct 13) → 57,535 (Tue Oct 20); 7-day retention of a day's new entrants 11–23 % for the weekday cohorts from Oct 7 (next day 6–10 %), ~30 % for the Oct 6 cohort (it still contains regulars who merely skipped the first day) and for the two small weekend cohorts (Fig. B1); revenue per active buyer-day $17, fees per active buyer-day averaging $47 ($36–65 by day).



Source: author's Solana collector, pump.fun/SOL, cohorts of buyers not seen earlier in the window (Oct 6–13; the Oct 5 cohort is excluded because on day 1 every buyer is 'new'); n = cohort size.

Fig. B1 — Seven-day retention of each day's new entrants on pump.fun (cohorts Oct 6–13; the Oct 6 cohort is inflated by regulars absent on day 1; the two weekend cohorts, Oct 10–11, are small and more loyal). Source: my Solana collector.

The food chain by hand (Fig. 4). Hands grouped by the number of positions taken in the 16 days, non-graduated tokens, strict P&L. The all-tokens variant (graduated included) gives −70 % for single-position hands and −3.7 % above 2,000 positions; it is biased against graduated tokens' buyers, whose sales happen on PumpSwap outside this data, so the body uses the non-graduated variant, consistent with Table 1 and Fig. 3.

Hands by positions in 16 days	Hands	% of hands	% of positions	% of SOL staked	Median strict P&L	Stake-weighted	% losing
1	507,199	44.9 %	3.2 %	3.2 %	–46.4 %	–19.8 %	84 %
2–5	437,462	38.7 %	8.7 %	6.0 %	–59.6 %	–24.1 %	84 %
6–20	115,276	10.2 %	8.3 %	6.8 %	–26.7 %	–18.7 %	79 %
21–100	53,404	4.7 %	17.1 %	15.1 %	–18.3 %	–16.0 %	75 %
101–500 (≈ 6–31 a day)	14,474	1.3 %	21.3 %	21.8 %	–13.4 %	–11.8 %	73 %
501–2,000 (≈ 31–125 a day)	1,452	0.1 %	9.7 %	9.9 %	–6.9 %	–7.9 %	69 %
> 2,000 (≥ 125 a day)	395	0.03 %	16.8 %	19.3 %	–4.0 %	–11.1 %	67 %

The "% of positions" and "% of SOL" columns are relative to the all-token totals (they sum to 85 % and 82 %); relative to the non-graduated totals, hands with five positions or fewer hold 14.0 % of positions and 11.2 % of the SOL, hands above 100 positions 56.2 % and 62.2 %. Script profile (more than 100 positions and active on at least 10 of the 16 days, all tokens; creators and snipers as scripts: Ding et al., ref. 17): 10,819 hands, 0.96 % of hands, 40.9 % of positions, 49.5 % of SOL staked. Occasional profile (five positions or fewer, all tokens): 944,661 hands, 83.6 % of hands, 15.4 % of positions, 13.4 % of SOL.

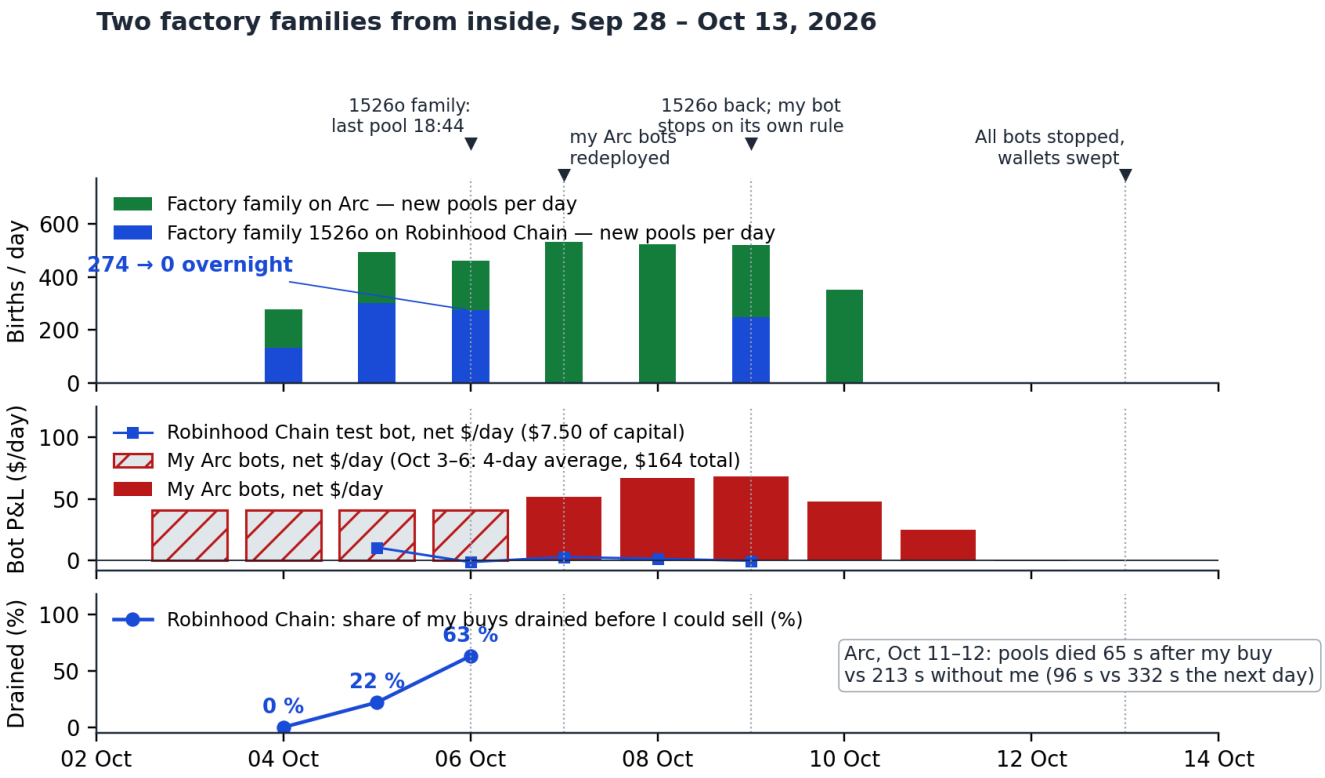
Holder map of PUMP (Solscan, October 22, 2026; top 80 accounts read with labels). 386,417 holders; top 10 = 60.3 % of total supply, top 100 = 88.6 %. By category: pump.fun custodian and allocation vaults 329.9B (39.9 % of total, ≈ the non-circulating supply); round-number Squads multisigs (19 accounts), 196.6B (42.6 % of circulating), several in identical tranches (3.125B ×3, 1.562B ×2); the same tranches also appear among unlabelled wallets (2.087B ×4, 3.125B ×1, 1.562B ×3) — read as unlocked insider allocations held, an inference from the amounts, not a Solscan label; exchanges and the Hyperliquid bridge 65.6B (14.2 %: Binance 27.4B, Kraken 11.1B, Hyperunit 9.0B, OKX 8.4B, Bybit 4.8B); unlabelled large wallets 118.5B (25.6 %); ranks 81 and beyond hold 113.5B (24.6 % of circulating). Raw table: `data/raw/solscan_pump_holders_2026-10-22.md` .

Appendix C — The hunt, day by day

Reconciliation of the money (on-chain where possible).

Item	Amount	Basis
External capital in, floor	$55.66 USDC on Arc, on or before Sep 23 (origin of this first deposit not reconstructed)	bot ledger
External capital in, ceiling	+ $10 (Sep 24) + $19.36 (Sep 25–29) + $0.90 (Sep 30) + $33.61 (Robinhood Chain, by Oct 2; bridged from Arc or from the vault, undetermined) + $2.93 (Base) = **$122.46**	lab notes
Balance trajectory (native balances, open positions excluded)	Sep 25 $40.81 → Oct 1 $201.24 → Oct 3 $330.29 → Oct 5 $510.42 → Oct 7 $579.50 → Oct 13 ≈ $804	balance snapshots
Recovered on Oct 13	640.77 USDC at the vault (four sweeps below = 629.14 USDC, plus 11.63 USDC from reserve wallets) + 0.042072 ETH (≈ $113 at $2,695/ETH) + 50 USDC converted to 0.176 TAO on Oct 8 (≈ $50) = **≈ $804**	on-chain
Campaign net, Sep 23 → Oct 13, four chains	$804 – $56 to $122 = **+$682 to +$748**	computed
Journals (one side only), Oct 3–12, Arc	≈ **+$424**; the balances over the same days went $330 → $804 (+$474): journals under-count by ≈ 10 %, which is why money is read on balances, not journals	journals vs balances
By chain	not separable exactly: the bridges Arc → Robinhood Chain (Relay) and Arc → BSC (Across) are internal; Arc carried ≈ 90 % of the P&L	—
September tests on Pons (Robinhood Chain)	≈ –0.26 ETH ≈ –$700 at $2,695/ETH (the September ETH price is not documented)	expense ledger
Overall, early September → Oct 13	**≈ –$20 to +$50 — about zero**	computed

53 addresses on Arc, Robinhood Chain, Base and BSC; the bots' parameters were unchanged between Oct 4 and 6 (journal). The "~$800" sometimes quoted is the final balance, not a gain.



Source: author's bot journals and lab reports, reconciled to on-chain balances (Appendix C); days without a reported value are left blank.

Fig. C1 — Two factory families from inside (Arc and Robinhood Chain): births per day (not logged after Oct 10), my bots' net dollars per day (journals, one side only; the Oct 3–12 entries sum to ≈ $424 of the $682–748 campaign), the share of my buys drained. Source: my collector and bot journals, Sep 28 – Oct 13, 2026.

Public proofs. The vault also served as the funding wallet of the campaign (its balance before the campaign is not part of the P&L). Sweep transactions on Arc (chain id 5042) to the vault 0x7d85bF7a82470837A1d832e4fa503a7ebF20ca97: - Oct 7, 2026 19:23 UTC — w000 → vault, 196.739407 USDC: `0x4278048243806f9effb75206445a33b6c29fb4519224220df66470b4b84c163a` - Oct 7, 2026 19:23 UTC — w001 → vault, 178.116421 USDC: `0x4c6db445c8349f39b5b3e40b4ea30339b2127f1a7d7bb4ce2b601abc0aef73ed` - Oct 13, 2026 21:56 UTC — f101 → vault, 143.872356 USDC: `0x99e5f0e954924dba3378e954fb8e67b2596e99d0de2edfe99093d4ce8812917e` - Oct 13, 2026 21:56 UTC — f102 → vault, 110.408233 USDC: `0x8dd41acced7790ea74812702f9fc50cd1ff9e6d6b149bd66f692074f1e27cced`

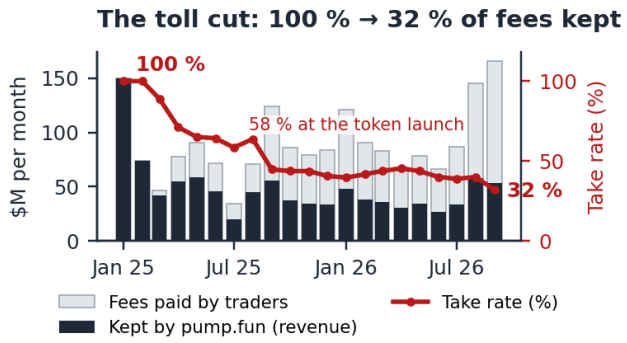
Addresses f101 and f102 can be found by a third party among the winning addresses of Arc's factory pools in the Oct 4–9 window from public logs alone (the Oct 9 lab report; over the full 17 days no address meets the "regular winner" rule of Appendix A). The day-by-day table (births per family, net dollars, pool life with and without my buy, share of my buys drained, events) is published with the data (`data/chasse-quotidien.csv`).

Appendix D — Valuation arithmetic

Target. Price = multiple × annual revenue ÷ circulating supply (461.9B). Multiple today = market cap $2.19B ÷ annualised trailing-30-day revenue $836M = 2.62×. Pre-mania base = Feb–Jun 2026 average $33.2M/month = $398M/yr. 2.62 × $398M = $1.04B ÷ 461.9B = **$0.00226 → $0.0022 (−54 %)**. The multiple has been pro-cyclical (1.9× at the June 2026 trough, 2.6× at the peak), so a constant multiple is the conservative assumption; the grid on page 4 shows the alternatives.

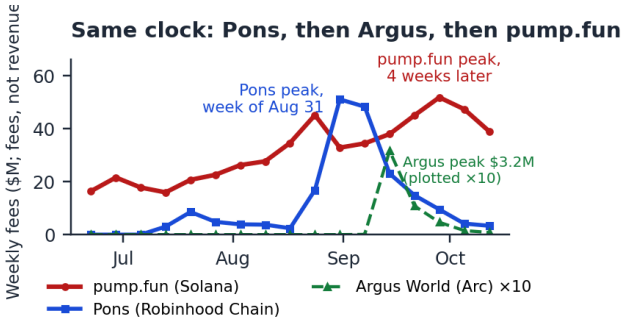
Reverse check (fully diluted). FDV $3.92B ÷ annual holder flow (50 % × $836M = $418M) = 9.4×. At a 25 % discount rate, r − g = 10.7 % → implied perpetual growth g ≈ 14 %/yr; the revenue that justifies today's FDV with no growth is R* = $3.92B × 0.25 ÷ 0.5 = $1.96B a year, forever ($1.10B on circulating market cap), against $836M for the best 30 days ever. Flat peak holder flows ($418M a year) are worth $418M ÷ 0.25 = FDV $1.67B (−57 %); base flows ($199M) $0.80B (−80 %).

Buyback-coverage price (no multiple). Net supply is zero when 50 % of annual revenue buys back exactly the annual insider issuance (82.5B tokens): P = 0.5 × revenue ÷ 82.5B. Run-rate $836M → $0.0051 (today's price is within 7 % of it); Feb–Jun base $398M → $0.0024; June trough $319M → $0.0019.



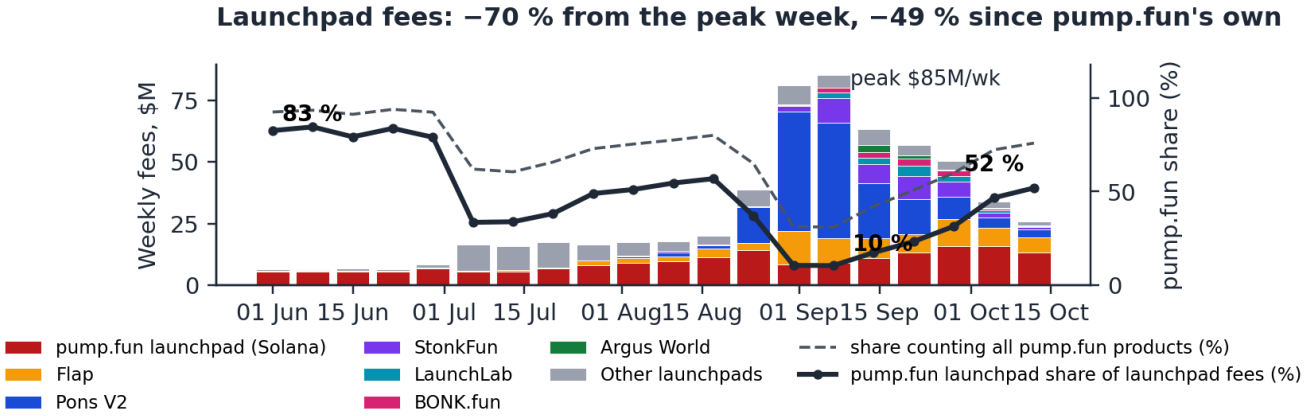
Source: DefiLlama dailyFees vs dailyRevenue, pump.fun parent, Jan 2025 – Sep 2026. Take rate = revenue / fees.

Fig. D1 — The take rate: 100 % → 58 % → 32 %.



Source: DefiLlama dailyFees, weeks of Jun 22 – Oct 12, 2026 (week of Oct 19 partial, excluded). Argus World scaled ×10 for visibility.

Fig. D2 —Three launchpads, one summer: weekly fees peaked in launch order. Source: DefiLlama.



Source: DefiLlama dailyFees, 16 launchpads with >$0.3M fees/30d in October (all chains), pump.fun counted as its launchpad only (dashed: share with PumpSwap, Terminal and app); ISO weeks 23–42 of 2026. Universe fixed on October fees (earlier casualties absent). Share = pump.fun / total.

Fig. D3 — Launchpad fees, launchpads only (16 venues, all chains; for pump.fun its launchpad, not PumpSwap, the Terminal or the app) and pump.fun's share; −70 % from the pie's peak week (Sep 7), −49 % since pump.fun's (Sep 28); dashed: pump.fun's share counting all its products (92 → 31 → 76 %). Pons is the largest slice of the peak week (57 %); universe fixed on October fees. Source: DefiLlama.

What the 2025 long said, and what I measure.

They said (the 2025 finalist long; the bull case today)	I measure	Where
85 % share of launches is a moat	83 % → 10 % → 52 % of launchpad fees in four months; the pie −70 %	Fig. D3
~100 % of revenue to buybacks; "$490M bought back"	50 %, by decision; 18¢ per fee dollar vs 58¢ at launch; unlocks outweigh buybacks at base revenue	Fig. D1
$870M of annualised fees, growing; target $0.0152	−87 % and −52 % from both prior peaks, −22 % from this one; over the last year the market paid 1.6× to 4.2× trailing revenue (month-ends), 1.9× at the June trough, 2.6× today	Fig. 1

Appendix E — Reproducibility

Every number recomputable from public data or from my published collector outputs has a command in the repository; the bot numbers (Fig. 2, Appendix C) are typed from my journals and reconciled to on-chain balances (GitHub, published before submission; numbers marked C in its README need the per-position files (3.9 GB) or the hourly collector files (8.6 GB), available on request): the raw API responses (`data/raw/` : DefiLlama, CoinGecko, Binance, Hyperliquid, the Solscan holder table), the 16 daily pump.fun results and their aggregates (`data/pumpfun-16d/`), the Arc 17-day ranking (`data/arc-17d/`), the daily bot table (`data/bots-daily.csv`), the lab tools (`scripts/lab/` , JavaScript, read-only, published; they run once per UTC day on the collector's hourly files, 8.6 GB compressed, available on request, as are the 3.9 GB of per-position files) and the figure

scripts (`scripts/`, Python 3, matplotlib). Cross-checks against Allium / Blockworks Research (active addresses, revenue, launches for Oct 5–20) will be added after the Oct 29 session; any gap above 10 % will be reported.